Nikhil Bhoj

Senior Software Engineer · Healthcare and Fintech

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Experience

Cedar Austin, TX

Senior Software Engineer *Nov 2021 – Present*

- Build and operate payment infrastructure for a healthcare billing platform serving 50M+ patients and processing $3.3B in patient payments annually ($10B+ lifetime).
- Architected Cedar's vendor-agnostic payment gateway, decoupling billing and collections flows from individual payment processors behind a uniform dispatch interface.
  - Shared base logic for charges, transfers, and refunds, with each processor independently implementable and testable in isolation.
  - Built Stripe as the first implementation on the abstraction; subsequent processors were added without changes to calling code.
- Optimized Stripe payment processing by eliminating redundant API calls across charge and refund paths, reducing payment processing costs by over $100K annually.
- Decomposed the payments monolith into event-driven microservices, improving throughput and enabling independent scaling of payment-critical workloads.
- Designed and delivered end-to-end integration with HealthEquity, Cedar's largest health benefit accounts partner.
  - Opened a new payment channel for health benefit accounts, driving millions of dollars in annual collections.
  - Handled eligibility, authorization, and settlement flows against the partner API, with reconciliation back into Cedar's billing records.
- Own the pre-visit patient experience platform; introduced ML-driven OCR for insurance card capture, accelerating eligibility verification and reducing manual intake overhead across health system clients.
- Built an ML-powered payer fax automation pipeline for prior authorization processing, reducing agent case time by up to 70%.
  - Ingested inbound payer faxes, converted them to images, and ran OCR to extract case fields.
  - Trained and served Naive Bayes models to detect and predict field values, pre-filling cases for agent review.
- Re-architected patient communications from blanket sends across all three channels to rules-based, per-client channel selection, eliminating redundant outreach and enabling targeted messaging strategies.

Actium Health (SymphonyRM) Austin, TX

Software Engineer *Jul 2019 – Nov 2021*

- Designed a high-throughput patient outreach platform on Twilio, built in Python and deployed as serverless services on AWS Lambda, powering bulk SMS campaigns across 14M patient lives.
  - Campaigns contributed to 1,395 early breast cancer diagnoses, recognized internally as a measurable clinical impact milestone.
- Built Apache Airflow ETL pipelines ingesting sensitive healthcare data from client SFTPs under HIPAA constraints.
  - Validated incoming client files, normalized records across inconsistent source schemas, and loaded them into production databases.

  - Partnered with data engineers and analysts to define ingestion contracts for new health system clients.

- Architected the standardized provider data export service powering external physician directories for all health system clients, processing millions of records daily.
- Automated new client provisioning, replacing a manual setup process and reducing onboarding time for new health system clients.
- Led a distributed offshore team of 5 engineers and served as scrum master, tracking milestones, unblocking teammates, and delivering cross-functional projects on schedule.
- Partnered with client success and sales teams on scoping, architecture, and estimation for customer requests and custom integrations.

Accenture Bengaluru, India
Software Engineer *Oct 2015 – Jul 2017*

- Built financial web applications in ReactJS and AngularJS for enterprise clients, owning features end-to-end from design through production deployment.
- Developed and consumed REST and SOAP web services connecting front-end applications to backend financial systems.
- Maintained customer relationship management databases in MySQL and PostgreSQL, writing queries and schema changes to support new application features.
- Worked directly with client stakeholders to triage production issues and ship enhancements on short turnarounds within an agile delivery process.

Education

Texas State University San Marcos, TX
M.S., Computer Science *2017 – 2019*

JSS Science and Technology University Mysuru, India
Master of Computer Applications *2012 – 2015*

University of Mysore Mysuru, India
Bachelor of Computer Applications *2009 – 2012*

Technical Skills

Languages: Python, JavaScript, TypeScript, C++
Frameworks: React, Flask, Django, Node.js, SQLAlchemy, gRPC, Serverless
Payments & Integrations: Stripe, PayPal, TrustCommerce, Twilio, TokenEx
Databases: PostgreSQL, MySQL, DynamoDB
Cloud & Infrastructure: AWS (Lambda), Docker, Kubernetes, Airflow, Harness, Datadog, Sentry
ML & Data: OCR, Naive Bayes, Anomaly Detection, ETL, Looker